Real-World Order of Operations

Question Bank - Additional 4.07 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, substitute and evaluate.

1. $5x^2$ when $x = -4$

80

$5(-4)^2 = 5(16)$.

2. $\frac{1}{2}mv^2$ when $m = 30$ and $v = 10$

1,500

$15(100)$.

3. $\frac{a^2 + b}{c}$ when $a = 8$, $b = 4$, $c = 17$

4

The bar groups the whole numerator: $64 + 4 = 68$.

4. $3(x + y)^2$ when $x = -5$ and $y = 2$

27

$3(-3)^2 = 3(9)$.

For questions 5 and 6, write and evaluate an expression. Give the unit.

5. A square room of side 13 feet is floored at 2.75 dollars a square foot, with a flat 90 dollars for delivery. Find the total cost.

$13^2(2.75) + 90 = 554.75$ ***dollars***

6. A rectangular patio 16 feet by 7 feet is sealed at 1.40 dollars a square foot, with a flat 35 dollars for equipment hire. Find the total cost.

$16(7)(1.40) + 35 = 191.80$ ***dollars***

The area is 112 square feet.

Two formulas.

7. Body mass index is $B = \frac{703w}{h^2}$, where w is a weight in pounds and h is a height in inches. Find B to the nearest tenth.

a. $w = 120$, $h = 62$

21.9

$\frac{84{,}360}{3{,}844} = 21.945\ldots$

b. $w = 165$, $h = 68$

25.1

$\frac{115{,}995}{4{,}624} = 25.085\ldots$

c. $w = 210$, $h = 74$

27.0

$\frac{147{,}630}{5{,}476} = 26.959\ldots$

8. Kinetic energy is $K = \frac{1}{2}mv^2$, for mass m in kilograms and speed v in meters per second. Find K in joules.

a. $m = 1{,}500$, $v = 20$

300,000 joules

b. $m = 640$, $v = 9$

25,920 joules

c. $m = 55$, $v = 4$

440 joules

Working backwards.

9. Find the missing value in each situation.

a. $\frac{1}{2}mv^2 = 1{,}600$ and $v = 8$. Find m.

m = 50

$32m = 1{,}600$.

b. $x^2 - 7 = 74$, where x is positive. Find x.

x = 9

c. A square lawn of side n feet is turfed at 3.00 dollars a square foot with a flat 55 dollars delivery, and the total is 1,255 dollars. Find n.

n = 20

$n^2(3.00) = 1{,}200$, *so* $n^2 = 400$.

Reading the situation.

10. Two problems use the numbers 8, 12 and 4. **A:** a printer costs 8 dollars to hire plus 12 dollars for each of 4 days. **B:** a printer costs 8 dollars for each of 12 reams, plus 4 dollars for delivery. Find both totals, and explain why circling the numbers would not have told you what to do.

A is $8+12(4)=56$ ***dollars. B is*** $8(12)+4=100$ ***dollars. The two problems carry the same three numbers, so the numbers cannot be what decides; what decides is which amount is repeated and how often.***

Ask which number is a rate in each problem — that question has a different answer in A and in B, and no marking on the page can show it.

11. A square court of side 20 feet is resurfaced at 3.60 dollars a square foot, with a flat 200 dollars for the equipment. Which expression gives the total cost?

- ○ A. $20^2(3.60+200)$
- ○ B. $20(3.60)+200$
- ● **C.** $20^2(3.60)+200$
- ○ D. $20^2+3.60+200$

C

12. Select ALL of the expressions whose value is 32 when $x=4$.

- ☒ **A.** x^2+16
- ☒ **B.** $\frac{128}{x}$
- ☐ C. $2x\cdot 2$
- ☒ **D.** $x^2\cdot 2$
- ☐ E. $(2x)^2$
- ☒ **F.** $2x^2$

A, B, D and F